UrbanNest Health Insurance Analysis: Findings & Recommendations

Simple Exploratory Data Analysis Report

Prepared by: Group 6
Role: Data Analyst / Data Scientist
Dataset: Medical Cost Personal Datasets (1,338 Records, 7 Fields)
Tools Used: Python (Pandas, Seaborn, Matplotlib, etc.), Jupyter Notebook, SQL.

Executive Summary

This report analyzes demographic and lifestyle factors affecting individual medical insurance charges. Using Python for exploratory data analysis (EDA), the study evaluates key drivers such as smoking status, Body Mass Index (BMI), age, region, and number of dependents.

1. Key Findings

From analyzing the health insurance dataset, here are the main patterns identified:

- **Smoking is the Biggest Cost Factor:** Smokers have much higher average medical charges than non-smokers ($32,050 vs. $8,434). Smoking is the single strongest indicator of higher costs in this dataset.

Group	Average Charge	Median Charge
Non-Smoker	$8,440	$7,345
Smoker	$32,050	$34,456

- **BMI Combined with Smoking Creates High Costs:** A high BMI alone increases costs slightly, but when combined with smoking, charges jump to over $41,000 on average.

- **Age Has a Steady Impact:** Charges increase steadily as age increases (roughly $250–$300 per year older), but it is a gradual rise compared to the sharp increase seen in smokers.

- **Demographics (sex, Region) Have Minimal Impact:** Average charges between male and female policyholders shows almost no difference with male having an average charge of $13,975 and female $12,569. Similarly, geographic region (Northeast, Southeast, Northwest, Southwest) shows negligible variations in all charges with southeast having a slightly higher average than the other regions but that isn't enough to say region is a key driver.

2. Recommendations

Based on these findings, here are the recommendations:

- **Focus on Smoking Cessation:** Because smokers account for the highest charges, creating smoking cessation programs offers the best opportunity to reduce overall claims.

- **Target High-BMI Smokers:** Health support programs should prioritize individuals who both smoke and have a high BMI, as they represent the highest financial risk group.

- **Adjust Premium Tiers:** Insurance pricing models should continue to account for smoking and BMI lifestyle factors to keep base rates fair for lower-risk groups.

- **Avoid Over-Segmenting by Region or Sex:** Since geography and gender show close to zero effect on cost differences, pricing structures should remain uniform across these categories rather than adding unnecessary complexity to premium calculations.